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Repository:
https://github.com/RabidInsidious/bayes460Project.git

Gaussian Naive Bayes Classification on the Bank Marketing Dataset

1. Dataset Review and Data Preprocessing

The Bank Marketing dataset, obtained from the UC Irvine Machine Learning Repository (Bank Marketing - UCI Machine Learning Repository), contains collected data from an unnamed Portuguese bank's direct marketing campaigns. The dataset consists of 45,211 observations and 17 columns, including customer information such as age, occupation, marital status, education, account balance, and previous marketing interactions. All these features were recorded to predict whether a customer will subscribe to a term deposit, represented by the binary target variable y (yes or no).

The original dataset contained semicolons as field separators; thus, the CSV file is read into the program before being altered and exported into a comma-separated file. Binary categorical values, such as "yes" and "no", were mapped to 1 and 0. The main challenge during task 1 was converting the non-binary categorical variables into compatible numerical values for the Gaussian Naive Bayes classifier. These variables, such as occupation, education, marital status, and contact type, were converted into integer representations starting at 0 using df[col] = pd. Categorical (df [col]) .codes.Categorical(df[col]).Rows containing missing values were removed using dropna().

Two main limitations of this approach are that the method of assigning integers to categorical variables creates artificial numerical relationships between categories that may not have any meaningful ordering. Additionally, values containing "unknown" are retained as valid categories rather than treated as missing data.

2. Experimental Objectives, Design, and Results

The goal of this experiment was to evaluate the performance of the Gaussian Naive Bayes Classifier and determine how changing the proportion of training data affects its accuracy and F1 score. The first portion of the experiment used a stratified random split of 70% training and 30% testing. The experiment was repeated on 10 iterations using random seeds, and the mean and standard deviation of both performance metrics were calculated.

The results of the model produced an average accuracy of **83.88%**, with a standard deviation of **0.00484**, and an average weighted F1 score of **84.75%**, with a standard deviation of **0.00379**. These small standard deviations indicate that the model's performance remained consistent across the different random splits.

The second experiment investigated the impact of training dataset size. Nine different training percentages were evaluated, increasing from 10% to 90% in increments of 10%. Each percentage was tested across 10 stratified random splits, resulting in 90 training and testing runs.

Training %	Mean Accuracy	Std. Accuracy	Mean F1	Std. F1
10%	0.8495	0.0117	0.8552	0.0072
20%	0.8426	0.0089	0.8501	0.0061
30%	0.8409	0.0080	0.8488	0.0058
40%	0.8428	0.0076	0.8503	0.0054
50%	0.8421	0.0073	0.8499	0.0053
60%	0.8394	0.0053	0.8481	0.0039
70%	0.8388	0.0048	0.8475	0.0038
80%	0.8388	0.0044	0.8476	0.0039
90%	0.8391	0.0044	0.8478	0.0036

An unexpected result was that the highest average accuracy, at **84.95%**, occurred with only 10% of the dataset used for training. Meaning increasing the training percentage did improve the classification's performance. Rather, the accuracy and weighted F1 gradually decreased before stabilizing around 83.9% and 84.8%, respectively, at larger training sizes. However, the standard deviations generally decreased as training size increased, implicating that using larger training sets produces consistent performance across random splits, despite not improving average classification scores.

3. Interpretation and Limitations

The results show that a Gaussian Naive Bayes model can produce stable predictions on the Bank Marketing dataset, but increasing the amount of training data does not improve performance. Likely due to the model reaching a point of diminishing returns. Another possibility is the underlying assumption of Gaussian Naive Bayes: the assumption that features are independent given the target class and that numerical features follow traditional Gaussian

distributions means that the model may not accurately represent the Bank Marketing dataset, particularly because several variables are potentially correlated.

The dataset is substantially disproportionate in favor of non-subscribed customers, meaning the accuracy might be misleading, since a model may achieve a high accuracy by correctly predicting the majority class while failing to identify customers who are actually subscribed. Although the F1 scores consider precision and recall, there might also be a greater influence of the majority class. Evaluating both true and false positives separately using solely precision, recall, and F1 might provide a clearer understanding of the model's ability to identify true potential subscribers. Additionally, the trend of decreasing standard deviations indicates that having a larger training dataset improves the results' consistency.

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